IP PROJECT

HOTEL MANAGEMENT SYSTEM

NAME: AARAV VASUDEV
CLASS: XII - C
SCHOOL: AMITY INTERNATIONAL SCHOOL SAKET

Table of Contents

- Certificate of Completion
- Acknowledgement
- Introduction
- Code
- Analysis

Certificate of Completion

This is to certify that the “Hotel Management System” project was completed by Aarav Vasudev of class XII-C from Amity International School, Saket under my supervision. He has taken proper care and shown utmost sincerity in completing the project. I certify that this project is up to my expectations and is adhering to the guidelines provided by CBSE.

Ms. Nidhi Anand
Informatics Practices Department
Amity International School, Saket

Acknowledgement

I would like to express my gratitude towards Ms. Nidhi Anand ma’am for guiding and supporting me throughout the completion of this project. Her inputs and suggestions were extremely valuable, and her feedback elevated the level of this project. Truly, this project wouldn’t have been a success without her.

Introduction

The “Hotel Management System” is a management software for allowing users to book Hotel rooms and maintain their records. The DMS is a menu-based application developed in Python (Spyder) with the backend of a comma-separated value (CSV) flat file.

The following menu options are available to the User:

1. Book their Hotel room.
2. Login/Register to confirm their bookings.
3. Viewing the rooms they have booked.
4. Cancelling the rooms they have booked.

If User is an Admin,

1. They can view all rooms booked by different people.
2. Viewing which Hotel / Room Type / Facilities is most popular using Graphs.
3. Managing User bookings and User data.

CSV Files

Storing User bookings and User Data

Bookings.csv

	A	B	C	D	E	F	G	H
1	Email	Hotel	Location	Room type	Room loc.	No. of People	Facilities	Price Range
2	aarav@gmail	Radisson Blu	USA	Suite	Poolside	1	Breakfast	Rs 0 - Rs 1500
3	exampleuser	Taj	Dubai	Deluxe	Beachside	1	All Facilities	Rs 5000 +
4	aryash	Pattaya Beach Resort	Thailand	Deluxe	Beachside	2	Buffet	Rs 2500 - Rs 5000
5	ryan	Radisson Blu	Singapore	Penthouse	Poolside	4	All Facilities	Rs 5000 +
6								

Users_data.csv

	A	B	C
1	email	password	
2	admin_user	admin12	
3	aarav@gmail	aarav123	
4	exampleuser	123	
5	aryash	1234	
6	ryan	123	
7			

Code

Code Analysis

Welcome Page

```
Welcome to ABC Hotel.
Choose a number:
1. Book rooms
2. View existing bookings
3. Cancel Bookings
4. Admin Login
How would you like to proceed?:
```

Finding Hotels

```
How would you like to proceed?: 1
Where would you like to stay for your holidays?
Please enter City, Property Name or Location: India
Please enter the Check-In Date (DD/MM): 03/02
Please enter the Check-Out Date (DD/MM): 09/02
Please enter the number of Rooms to be booked: 1
Please enter the number of Guests staying in the room: 1
Please choose a Price Range:
1. Rs 0 - Rs 1500
2. Rs 1500 - Rs 2500
3. Rs 2500 - Rs 5000
4. Rs 5000 +
Choose a number: 3
Looking for rooms under the Price Range  Rs 2500 - Rs 5000 ...
Rooms found!
----------------------
Showing Properties in  India under the Price Range Rs 2500 - Rs 5000
1. Radisson Blu
- Rating: Excellent [5 stars]
2. The Imperial
- Rating: Excellent [4.5 stars]
3. Taj
- Rating: Very Good [4 stars]
4. Hard Rock Hotel
- Rating: Very Good [3.8 stars]
5. Pattaya Beach Resort
- Rating: Very Good [4 stars]
----------------------
Please choose a number: 4
Hotel Chosen: Hard Rock Hotel
```

Code Analysis

Choosing Room Types and Facilities

```
Hotel Chosen: Hard Rock Hotel
Booking a room!
Choose a room type:
1. Suite
2. Executive Suite
3. Deluxe
4. Villa
5. Penthouse
Choose a number: 4
Villa selected.
Rooms available!
----------------------
Choose your room location:
1. Poolside
2. Beachside
Choose a number: 1
Poolside selected.
Room available!
----------------------
Choose facilities to be included:
1. Breakfast included
2. Buffet
3. Gym membership
4. All
5. None
Choose a number: 3
Gym membership selected.
----------------------
Confirming your bookings!
Confirm? (Y/N):
```

Registering (Email already exists)

```
Ok! To confirm your booking, you may have to log in first!
1. Register
2. Login
3. Exit
Choose a number: 1
Enter email: ryan
A user with that email already exists. Please use a different email.
Do you want to log in instead? (Y/N): n
Please choose a different email.
Enter new email: ryan12
Enter Password: 123
Registered successfully!
You may have to Log in now...
Enter email: ryan12
Enter Password: 123
Login successful!
Here is your booking:
    Email           Hotel Location Room type Room loc.  No. of People     Facilities       Price Range
4  ryan12  Hard Rock Hotel    India     Villa  Poolside             1  Gym Membership  Rs 2500 - Rs 5000
```

Code Analysis

Registering (New email)

```
Ok! To confirm your booking, you may have to log in first!
1. Register
2. Login
3. Exit
Choose a number: 1
Enter email: shivam12
Enter password: guitar
Registered successfully!
You may have to Log in now...
Enter email: shivam12
Enter Password: guitar
Login successful!
Here is your booking:
      Email           Hotel Location Room type Room loc.  No. of People      Facilities        Price Range
5  shivam12  Hard Rock Hotel    India     Villa  Poolside             1  Gym Membership  Rs 2500 - Rs 5000
```

Login (Shows your recent bookings despite having more than 1)

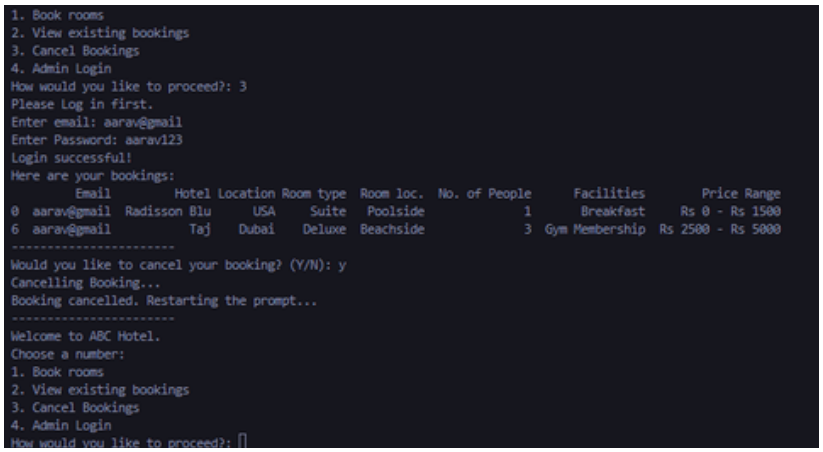
```
Ok! To confirm your booking, you may have to log in first!
1. Register
2. Login
3. Exit
Choose a number: 2
Enter email: aarav@gmail
Enter Password: aarav123
Login successful!
Here is your booking:
        Email Hotel Location Room type  Room loc.  No. of People      Facilities        Price Range
6  aarav@gmail   Taj    Dubai    Deluxe  Beachside             3  Gym Membership  Rs 2500 - Rs 5000
Moving on to Payment. Kindl Pay using UPI or Pay at Reception.
Thank you for booking with us! Have a nice day
```

Viewing your previous bookings

```
Welcome to ABC Hotel.
Choose a number:
1. Book rooms
2. View existing bookings
3. Cancel Bookings
4. Admin Login
How would you like to proceed?: 2
You may have to log in first!
Enter email: aarav@gmail
Enter Password: aarav123
Login successful!
        Email        Hotel Location Room type  Room loc.  No. of People      Facilities        Price Range
0  aarav@gmail  Radisson Blu      USA     Suite   Poolside             1       Breakfast     Rs 0 - Rs 1500
6  aarav@gmail          Taj    Dubai    Deluxe  Beachside             3  Gym Membership  Rs 2500 - Rs 5000
```

Code Analysis

Cancel Bookings (Cancels most recent booking)



```
1. Book rooms
2. View existing bookings
3. Cancel Bookings
4. Admin Login
How would you like to proceed?: 3
Please Log in first.
Enter email: aarav@gmail
Enter Password: aarav123
Login successful!
Here are your bookings:
        Email        Hotel Location Room type  Room loc.  No. of People      Facilities       Price Range
0  aarav@gmail  Radisson Blu      USA     Suite   Poolside             1       Breakfast     Rs 0 - Rs 1500
6  aarav@gmail          Taj    Dubai    Deluxe  Beachside             3  Gym Membership  Rs 2500 - Rs 5000
----------------------
Would you like to cancel your booking? (Y/N): y
Cancelling Booking...
Booking cancelled. Restarting the prompt...
----------------------
Welcome to ABC Hotel.
Choose a number:
1. Book rooms
2. View existing bookings
3. Cancel Bookings
4. Admin Login
How would you like to proceed?:
```

Email	Hotel	Location	Room type	Roo
aarav@gmail	Radisson Blu	USA	Suite	Poo
exampleuser	Taj	Dubai	Deluxe	Bea
aryash	Pattaya Beach Resort	Thailand	Deluxe	Bea
ryan	Radisson Blu	Singapore	Penthouse	Poo
shivam12	Hard Rock Hotel	India	Villa	Poo
aarav@gmail	Radisson Blu	USA	Suite	Poo

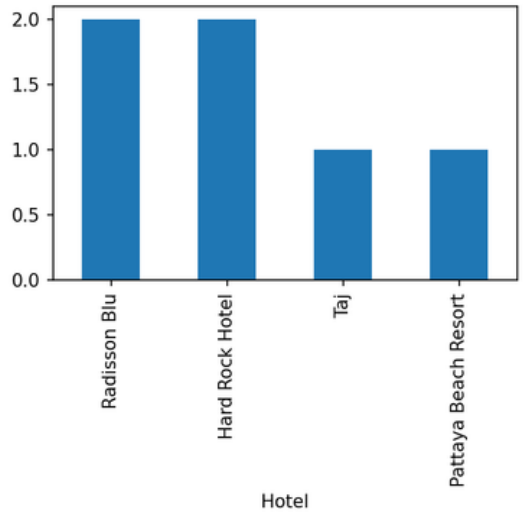
Email	Hotel	Location	Room type	F
aarav@gmail	Radisson Blu	USA	Suite	F
exampleuser	Taj	Dubai	Deluxe	B
aryash	Pattaya Beach Resort	Thailand	Deluxe	B
ryan	Radisson Blu	Singapore	Penthouse	F
shivam12	Hard Rock Hotel	India	Villa	F

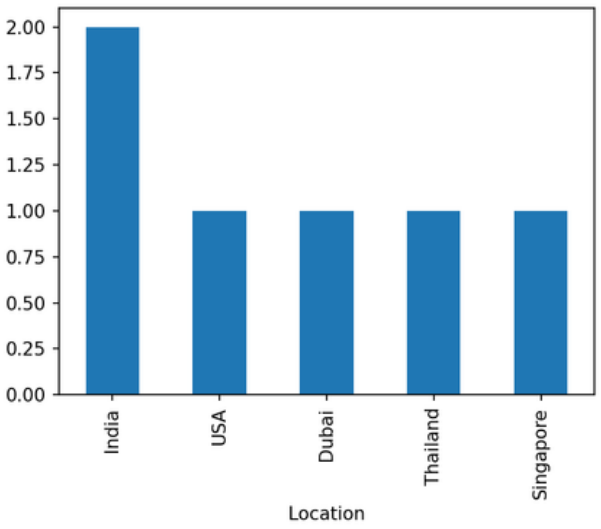
Admin Login (View all bookings)

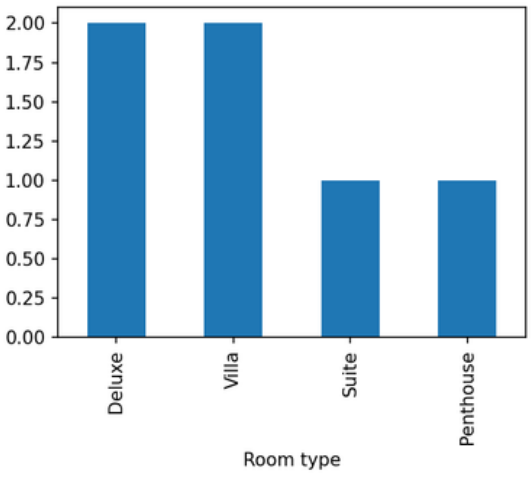
```
Choose a number:
1. Book rooms
2. View existing bookings
3. Cancel Bookings
4. Admin Login
How would you like to proceed?: 4
Enter email: admin_user
Enter Password: admin12
Login successful!
        Email               Hotel   Location  Room type  Room loc.  No. of People      Facilities       Price Range
0  aarav@gmail         Radisson Blu       USA      Suite   Poolside             1       Breakfast     Rs 0 - Rs 1500
1  exampleuser                 Taj     Dubai     Deluxe  Beachside             1  All Facilities          Rs 5000 +
2       aryash  Pattaya Beach Resort   Thailand     Deluxe  Beachside             2         Buffet  Rs 2500 - Rs 5000
3         ryan         Radisson Blu  Singapore  Penthouse   Poolside             4  All Facilities          Rs 5000 +
4      ryan12      Hard Rock Hotel      India      Villa   Poolside             1  Gym Membership  Rs 2500 - Rs 5000
5     shivam12      Hard Rock Hotel      India      Villa   Poolside             1  Gym Membership  Rs 2500 - Rs 5000
----------------------
1. Most Popular Hotels.
2. Most Popular Location
3. Most Popular Room type.
4. Most Popular Facility chosen.
5. Manage Bookings.
6. Manage Users.
Choose a number:
```

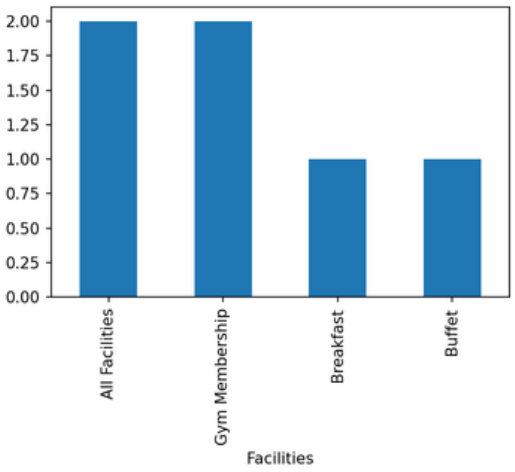
Code Analysis

Admin (Most Popular Commodities)









Code Analysis

Admin (Managing Bookings)
Add a new booking

```
----------------------
1. Most Popular Hotels.
2. Most Popular Location
3. Most Popular Room type.
4. Most Popular Facility chosen.
5. Manage Bookings.
6. Manage Users.
Choose a number: 5
        Email               Hotel   Location  Room type  Room loc.  No. of People      Facilities        Price Range
0  aarav@gmail         Radisson Blu       USA      Suite   Poolside              1       Breakfast     Rs 0 - Rs 1500
1  exampleuser                 Taj     Dubai     Deluxe  Beachside              1  All Facilities          Rs 5000 +
2       aryash  Pattaya Beach Resort   Thailand     Deluxe  Beachside              2          Buffet  Rs 2500 - Rs 5000
3         ryan         Radisson Blu  Singapore  Penthouse   Poolside              4  All Facilities          Rs 5000 +
4       ryan12      Hard Rock Hotel      India      Villa   Poolside              1  Gym Membership  Rs 2500 - Rs 5000
5     shivam12      Hard Rock Hotel      India      Villa   Poolside              1  Gym Membership  Rs 2500 - Rs 5000
----------------------
1. Add a new booking
2. Delete a booking
Choose a number: 1
Please Start the Prompt from the beginning and Book a room to add new bookings
```

Admin (Managing Bookings)
Deleting a booking

```
----------------------
1. Most Popular Hotels.
2. Most Popular Location
3. Most Popular Room type.
4. Most Popular Facility chosen.
5. Manage Bookings.
6. Manage Users.
Choose a number: 5
        Email               Hotel   Location  Room type  Room loc.  No. of People      Facilities        Price Range
0  aarav@gmail         Radisson Blu       USA      Suite   Poolside              1       Breakfast     Rs 0 - Rs 1500
1  exampleuser                 Taj     Dubai     Deluxe  Beachside              1  All Facilities          Rs 5000 +
2       aryash  Pattaya Beach Resort   Thailand     Deluxe  Beachside              2          Buffet  Rs 2500 - Rs 5000
3         ryan         Radisson Blu  Singapore  Penthouse   Poolside              4  All Facilities          Rs 5000 +
4       ryan12      Hard Rock Hotel      India      Villa   Poolside              1  Gym Membership  Rs 2500 - Rs 5000
5     shivam12      Hard Rock Hotel      India      Villa   Poolside              1  Gym Membership  Rs 2500 - Rs 5000
----------------------
1. Add a new booking
2. Delete a booking
Choose a number: 2
Please enter the email of the booking you want to cancel (Case sensitive): ryan12
Ok
User Bookings Updated.
```

Code Analysis

Admin (Managing Bookings)
CSV Before and After deleting bookings

[Before deleting ryan12]

Email	Hotel	Location	Room type	Room loc.	No. of People	Facilities	Price Range
aarav@gmail	Radisson Blu	USA	Suite	Poolside	1	Breakfast	Rs 0 - Rs 1500
exampleuser	Taj	Dubai	Deluxe	Beachside	1	All Facilities	Rs 5000 +
aryash	Pattaya Beach Resort	Thailand	Deluxe	Beachside	2	Buffet	Rs 2500 - Rs 5000
ryan	Radisson Blu	Singapore	Penthouse	Poolside	4	All Facilities	Rs 5000 +
ryan12	Hard Rock Hotel	India	Villa	Poolside	1	Gym Membership	Rs 2500 - Rs 5000
shivam12	Hard Rock Hotel	India	Villa	Poolside	1	Gym Membership	Rs 2500 - Rs 5000

[After deleting ryan12]

Email	Hotel	Location	Room type	Room loc.	No. of People	Facilities	Price Range
aarav@gmail	Radisson Blu	USA	Suite	Poolside	1	Breakfast	Rs 0 - Rs 1500
exampleuser	Taj	Dubai	Deluxe	Beachside	1	All Facilities	Rs 5000 +
aryash	Pattaya Beach Resort	Thailand	Deluxe	Beachside	2	Buffet	Rs 2500 - Rs 5000
ryan	Radisson Blu	Singapore	Penthouse	Poolside	4	All Facilities	Rs 5000 +
shivam12	Hard Rock Hotel	India	Villa	Poolside	1	Gym Membership	Rs 2500 - Rs 5000

Code Analysis

Admin (Managing Users)
Adding a User

```
----------------------
1. Most Popular Hotels.
2. Most Popular Location
3. Most Popular Room type.
4. Most Popular Facility chosen.
5. Manage Bookings.
6. Manage Users.
Choose a number: 6
        email  password
0   admin_user   admin12
1  aarav@gmail  aarav123
2  exampleuser       123
3       aryash      1234
4         ryan       123
5       ryan12       123
6     shivam12    guitar
----------------------
1. Add a new User
2. Delete a User
Choose a number: 1
Adding a new user...
Please enter email: abc
Please enter password: 123
User successfully added! Please restart the prompt to Refresh the page.
```

Admin (Managing Users)
Deleting a User

```
----------------------
1. Most Popular Hotels.
2. Most Popular Location
3. Most Popular Room type.
4. Most Popular Facility chosen.
5. Manage Bookings.
6. Manage Users.
Choose a number: 6
        email  password
0   admin_user   admin12
1  aarav@gmail  aarav123
2  exampleuser       123
3       aryash      1234
4         ryan       123
5       ryan12       123
6     shivam12    guitar
7          abc       123
----------------------
1. Add a new User
2. Delete a User
Choose a number: 2
Deleting a user...
Please enter the email of the booking you want to cancel (Case sensitive): abc
Ok
Users Updated.
User successfully added! Please restart the prompt to Refresh the page.
----------------------
```

Code Analysis

Admin (Managing Users)
CSV Before and After managing users

[Adding ‘abc’]

email	password
admin_user	admin12
aarav@gmail	aarav123
exampleuser	123
aryash	1234
ryan	123
ryan12	123
shivam12	guitar
abc	123

[Deleting ‘abc’]

email	password
admin_user	admin12
aarav@gmail	aarav123
exampleuser	123
aryash	1234
ryan	123
ryan12	123
shivam12	guitar